Factory Acceptance Test Procedure - Control and Safety Systems

DEMO-GPF Gas Processing Facility (fictional)
Fictional Operator Company (FOC)

STATUS: DRAFT - AI-ASSISTED, NOT CHECKED. Not for use until approved by a named engineer.

Document No.	Revision	Status	Generated
GPF-DEMO-PRO-001	A-draft	Draft	2026-10-10

Role	Name	Signature	Date
Prepared			
Checked			
Approved			

Approval is by named, competent engineers only. The copilot never signs.

1 Purpose

Placeholder - narrative not yet drafted
[NARRATIVE TO BE DRAFTED - purpose] Why this FAT is performed and what acceptance it gives (2-4 sentences).

2 Scope

Placeholder - narrative not yet drafted
[NARRATIVE TO BE DRAFTED - scope] Systems, cabinets, units and what is excluded (2-4 sentences).

3 Reference documents

Document No.	Rev	Title / dataset	File
GPF-IC-LST-003	A	alarm list	data/GPF-IC-LST-003_Alarm_Trip_List.xlsx
GPF-SS-LST-002	A	bms sequence	data/GPF-SS-LST-002_BMS_Sequence.xlsx
GPF-SS-CE-001	A	esd cause effect	data/GPF-SS-CE-001_ESD_Cause_and_Effect.xlsx
GPF-SS-CE-002	A	fg cause effect	data/GPF-SS-CE-002_FG_Cause_and_Effect.xlsx
GPF-IC-LST-001	B	instrument index	data/GPF-IC-LST-001_Instrument_Index.xlsx
GPF-IC-LST-002	B	io list	data/GPF-IC-LST-002_IO_List.xlsx
GPF-SS-LST-001	A	sif register	data/GPF-SS-LST-001_SIF_Register.xlsx

4 Prerequisites

☐ Hardware fully assembled, wired and internally checked by the vendor; inspection records available.

☐ Application software loaded at the revision stated in the test record; version recorded.

☐ Power-on and system diagnostics healthy; no active system alarms.

☐ Test equipment calibrated and within calibration date; certificates available.

☐ Latest revisions of the I/O list, cause & effect matrix and alarm list available at the test location.

☐ Open consistency-check findings reviewed (see section 6); none affects the scope under test.

5 Test equipment

☐ mA / HART calibrator (ID and calibration date recorded)

☐ Decade resistance box for RTD simulation

☐ Digital multimeter

☐ Test leads and simulation switches for discrete inputs

6 Data status at generation

The latest consistency check (20261010T152443Z-4de462) reported 39 open findings. Tests touching these items are marked; resolve at source before the FAT where they affect the scope.

Severity	Rule	Item	Finding
critical	IC-012	300-PT-040A	Trip setpoint for SIF-900-06 differs between documents: SIF register 150 barg, C&E 148 barg, alarm list 150 barg
critical	IC-012	300-PT-040B	Trip setpoint for SIF-900-06 differs between documents: SIF register 150 barg, C&E 148 barg, alarm list 150 barg
critical	IC-014	200-LT-031	System segregation: system: instrument 'SIS' vs io_point 'BPCS'
critical	IC-014	800-GD-101C	System segregation: system: instrument 'FGS' vs io_point 'SIS'
critical	IC-015	100-PT-005	Ex ia device allocated to non-IS channel CAB-BPCS-01/R1/S02/CH02 (AI-HART)
critical	IC-017	200-BDV-043	C&E effect references 200-BDV-043, which is not in the instrument index
critical	IC-020	SIF-900-04	SIL 2 SIF uses 1oo1 initiators (HFT 0); minimum HFT 1 required
critical	IC-022	100-TT-007	certified gas group IIA does not cover area group IIB
critical	IC-022	400-BE-020B	certified temperature class T2 does not cover area T3
critical	IC-025	300-PT-040B	Redundant devices 300-PT-040A, 300-PT-040B share card CAB-SIS-01/R1/S01
critical	IC-025	400-BE-020B	Redundant devices 400-BE-020A, 400-BE-020B share card CAB-SIS-01/R1/S05
critical	IC-026	300-PT-040B	C&E cause for 300-PT-040B (SIF-900-06) does not action final element 300-ESDV-042
critical	IC-027	300-PT-044A	C&E voting 2oo2 differs from SIF register 1oo2 for SIF-900-07
critical	IC-027	300-PT-044B	C&E voting 2oo2 differs from SIF register 1oo2 for SIF-900-07
critical	IC-030	400-ZSC-204	Sequence step references 400-ZSC-204, which is not in the instrument index
critical	IC-031	BMS-400 step 01	PURGE time 120 s is below the minimum 300 s
critical	IC-031	BMS-400 step 03	PILOT_TFI time 15 s exceeds the maximum 10 s
critical	IC-032	400-XV-023	C&E effect E12 action 'CLOSE' does not match the safe state 'OPEN' in the instrument index (expected 'OPEN')
critical	IC-032	700-BDV-012	C&E effect E15 action 'OPEN' does not match the safe state 'CLOSED' in the instrument index (expected 'CLOSE')
critical	IC-035	FG-300-FIRE	Voting 1oo3 drives executive actions (300-ESDV-041, 300-ESDV-042, 300-XY-043, 700-BDV-013); at least 2ooN required
critical	IC-036	FG-100-H2S-H	Voting 2oo3 but only 2 initiator(s) listed in the F&G C&E
major	IC-002	200-LT-012	Tag '200-LT-012' appears 2 times (service: Separator condensate level, Separator condensate level (local indication))
major	IC-003	CAB-BPCS-02/R1/S01/CH06	I/O address 'CAB-BPCS-02/R1/S01/CH06' appears 2 times (tag: 200-PT-021, 200-TT-017)
major	IC-005	300-TT-005	300-TT-005 has a signal type in the instrument index but no channel in the I/O list
major	IC-007	200-PT-011	Range mismatch: urv: instrument '100' vs io_point '60'
major	IC-008	100-PT-003	Unit mismatch: eu: instrument 'barg' vs io_point 'bara'
major	IC-010	200-TT-015	H setpoint 165 degC is outside the instrument range 0–150 degC
major	IC-010	300-VT-031	HH setpoint 160 µm is outside the instrument range 0–150 µm
major	IC-010	600-AT-010	H setpoint 60 ppm is outside the instrument range 0–50 ppm
major	IC-011	200-PT-021	HH setpoint 20 is not above H setpoint 21 (trip setpoint inside the alarm band)
major	IC-013	200-TT-016	signal 'RTD Pt100' is not compatible with card_type 'AI-HART-IS' (allowed: 4-20mA, 4-20mA HART)
major	IC-019	CAB-BPCS-02	3 spare of 24 installed channels (12%); minimum is 20%
major	IC-028	300-TT-021	eu 'barg' is not compatible with instrument_type 'TT' (allowed: degC)
major	IC-029	300-TT-021	Alarm unit mismatch: eu: alarm 'degC' vs instrument 'barg'
major	IC-034	800-GD-101B	H setpoint 60 %LEL differs from the standard 50 %LEL for GD
major	IC-037	500-FT-010B	Meter run B ultrasonic flowmeter: companion instrument(s) 500-TT-010B not in the instrument index
minor	IC-001	100-PT020	tag '100-PT020' does not match the convention \d{3}-[A-Z]{1,4}-\d{3}[A-Z]?
minor	IC-021	200-FT-018	Transmitter is not HART-enabled (hart = 0)
minor	IC-024	300-PT-020	L setpoint 0.5 barg is at 5% of span (0–10); preferred band 10%–90%

7 I/O test (100 %)

Each wired channel is tested from the marshalling terminals to the operator display. Spare channels are checked for availability per cabinet.

CAB-BPCS-01 (BPCS) - 13 channels

Tag	Address	Card	Test	Expected	Result	Init.
100-PT-001	R1/S01/CH01	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 75 / 150 barg		
100-PT-003	R1/S01/CH02	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 75 / 150 barg		
100-TT-007	R1/S01/CH04	AI-HART-IS	Inject 4 / 12 / 20 mA	-20 / 30 / 80 degC		
100-LT-008	R1/S01/CH05	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 %		
100-FT-010	R1/S01/CH06	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 200 / 400 kSm3/h		
100-PDT-015	R1/S01/CH07	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 250 / 500 mbar		
100-TT-016	R1/S01/CH08	AI-HART-IS	Inject 4 / 12 / 20 mA	-20 / 30 / 80 degC		
100-PT020	R1/S01/CH09	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 75 / 150 barg		
100-AT-017	R1/S02/CH01	AI-HART	Inject 4 / 12 / 20 mA	0 / 25 / 50 ppm		
100-PT-005	R1/S02/CH02	AI-HART	Inject 4 / 12 / 20 mA	0 / 75 / 150 barg		
100-TE-018	R1/S03/CH01	RTD-IS	Simulate 0 / 50 / 100 %	-20 / 30 / 80 degC		
100-PV-004	R1/S04/CH01	AO-HART-IS	Command 0 / 50 / 100 %	4 / 12 / 20 mA at terminals		
100-LV-008	R1/S04/CH02	AO-HART-IS	Command 0 / 50 / 100 %	4 / 12 / 20 mA at terminals		

CAB-BPCS-02 (BPCS) - 22 channels

Tag	Address	Card	Test	Expected	Result	Init.
200-PT-011	R1/S01/CH01	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 barg		
200-LT-012	R1/S01/CH02	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 %		
200-LT-013	R1/S01/CH03	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 %		
200-TT-015	R1/S01/CH04	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 75 / 150 degC		
200-TT-016	R1/S01/CH05	AI-HART-IS	Simulate 0 / 50 / 100 %	0 / 125 / 250 degC		
200-TT-017	R1/S01/CH06	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 75 / 150 degC		
200-PT-021	R1/S01/CH06	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 12.5 / 25 barg		
200-FT-018	R1/S01/CH07	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 25 / 50 m3/h		
200-LT-031	R1/S01/CH08	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 %		
200-PT-022	R1/S01/CH09	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 12.5 / 25 barg		
200-TT-023	R1/S01/CH10	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 125 / 250 degC		
200-LT-024	R1/S01/CH11	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 %		
200-FT-025	R1/S01/CH12	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 25 / 50 m3/h		
200-PDT-026	R1/S01/CH13	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 250 / 500 mbar		
200-TT-027	R1/S01/CH14	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 degC		
200-PV-011	R1/S02/CH01	AO-HART-IS	Command 0 / 50 / 100 %	4 / 12 / 20 mA at terminals		
200-LV-012	R1/S02/CH02	AO-HART-IS	Command 0 / 50 / 100 %	4 / 12 / 20 mA at terminals		
200-LV-013	R1/S02/CH03	AO-HART-IS	Command 0 / 50 / 100 %	4 / 12 / 20 mA at terminals		
200-PV-021	R1/S02/CH04	AO-HART-IS	Command 0 / 50 / 100 %	4 / 12 / 20 mA at terminals		
200-TV-023	R1/S02/CH05	AO-HART-IS	Command 0 / 50 / 100 %	4 / 12 / 20 mA at terminals		
200-LV-024	R1/S02/CH06	AO-HART-IS	Command 0 / 50 / 100 %	4 / 12 / 20 mA at terminals		
200-FV-025	R1/S02/CH07	AO-HART-IS	Command 0 / 50 / 100 %	4 / 12 / 20 mA at terminals		

CAB-BPCS-03 (BPCS) - 21 channels

Tag	Address	Card	Test	Expected	Result	Init.
300-PT-001	R1/S01/CH01	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 barg		
300-TT-002	R1/S01/CH02	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 degC		
300-FT-003	R1/S01/CH03	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 200 / 400 kSm3/h		
300-PT-004	R1/S01/CH04	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 80 / 160 barg		
300-PDT-006	R1/S01/CH06	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 250 / 500 mbar		
300-TT-007	R1/S01/CH07	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 degC		
300-FT-011	R1/S01/CH08	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 25 / 50 Nm3/h		
300-PDT-012	R1/S01/CH09	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 500 / 1000 mbar		
300-PT-013	R1/S01/CH10	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 barg		

Tag	Address	Card	Test	Expected	Result	Init.
300-PT-020	R1/S01/CH11	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 5 / 10 barg		
300-TT-021	R1/S01/CH12	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 barg		
300-LT-022	R1/S01/CH13	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 %		
300-VT-030	R1/S02/CH01	AI-HART	Inject 4 / 12 / 20 mA	0 / 75 / 150 µm		
300-VT-031	R1/S02/CH02	AI-HART	Inject 4 / 12 / 20 mA	0 / 75 / 150 µm		
300-VT-032	R1/S02/CH03	AI-HART	Inject 4 / 12 / 20 mA	0 / 75 / 150 µm		
300-VT-033	R1/S02/CH04	AI-HART	Inject 4 / 12 / 20 mA	0 / 75 / 150 µm		
300-ZT-034A	R1/S02/CH05	AI-HART	Inject 4 / 12 / 20 mA	-1 / 0 / 1 mm		
300-ST-036	R1/S02/CH06	AI-HART	Inject 4 / 12 / 20 mA	0 / 6000 / 12000 rpm		
300-ZT-034B	R1/S03/CH01	AI-HART	Inject 4 / 12 / 20 mA	-1 / 0 / 1 mm		
300-TE-037	R1/S04/CH01	RTD-IS	Simulate 0 / 50 / 100 %	0 / 75 / 150 degC		
300-FV-003	R1/S05/CH01	AO-HART-IS	Command 0 / 50 / 100 %	4 / 12 / 20 mA at terminals		

CAB-BPCS-04 (BPCS) - 14 channels

Tag	Address	Card	Test	Expected	Result	Init.
400-PT-001	R1/S01/CH01	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 barg		
400-PDT-002	R1/S01/CH02	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 250 / 500 mbar		
400-LT-003	R1/S01/CH03	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 %		
400-FT-005	R1/S01/CH04	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 2.5 / 5 m3/h		
400-TT-006	R1/S01/CH05	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 degC		
400-TT-010	R1/S01/CH06	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 125 / 250 degC		
400-LT-011	R1/S01/CH07	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 %		
400-PT-012	R1/S01/CH08	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 5 / 10 barg		
400-TT-013	R1/S01/CH09	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 300 / 600 degC		
400-PT-014	R1/S01/CH10	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 5 / 10 barg		
400-AT-004	R1/S02/CH01	AI-HART	Inject 4 / 12 / 20 mA	0 / 100 / 200 mg/Sm3		
400-LV-003	R1/S03/CH01	AO-HART-IS	Command 0 / 50 / 100 %	4 / 12 / 20 mA at terminals		
400-FV-005	R1/S03/CH02	AO-HART-IS	Command 0 / 50 / 100 %	4 / 12 / 20 mA at terminals		
400-TV-010	R1/S03/CH03	AO-HART-IS	Command 0 / 50 / 100 %	4 / 12 / 20 mA at terminals		

CAB-BPCS-05 (BPCS) - 20 channels

Tag	Address	Card	Test	Expected	Result	Init.
500-PT-001	R1/S01/CH01	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 80 / 160 barg		
600-PT-001	R1/S01/CH02	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 30 / 60 barg		
600-FT-002	R1/S01/CH03	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 25 / 50 m3/h		
600-FT-011	R1/S01/CH04	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 15 / 30 m3/h		
600-LT-012	R1/S01/CH05	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 %		
700-LT-001	R1/S01/CH06	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 %		
700-PT-021	R1/S01/CH07	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 2.5 / 5 barg		
700-FT-022	R1/S01/CH08	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 200 / 400 kSm3/h		
700-TT-020A	R1/S01/CH09	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 600 / 1200 degC		
600-AT-010	R1/S02/CH01	AI-HART	Inject 4 / 12 / 20 mA	0 / 25 / 50 ppm		
700-TT-020B	R1/S03/CH01	AI-HART-IS	Inject 4 / 12 / 20 mA	0 / 600 / 1200 degC		
600-LV-012	R1/S04/CH01	AO-HART-IS	Command 0 / 50 / 100 %	4 / 12 / 20 mA at terminals		
500-ZSC-011A	R1/S05/CH01	DI-NAMUR-IS	Simulate contact / NAMUR state	State change at HMI		
500-ZSC-011B	R1/S06/CH01	DI-NAMUR-IS	Simulate contact / NAMUR state	State change at HMI		
600-YS-001A	R1/S07/CH01	DI-24V	Simulate contact / NAMUR state	State change at HMI		
600-YS-001B	R1/S08/CH01	DI-24V	Simulate contact / NAMUR state	State change at HMI		
500-XV-011A	R1/S09/CH01	DO-24V	Force output on / off	24 VDC on / off at terminals		
600-YC-001A	R1/S09/CH02	DO-24V	Force output on / off	24 VDC on / off at terminals		
500-XV-011B	R1/S10/CH01	DO-24V	Force output on / off	24 VDC on / off at terminals		
600-YC-001B	R1/S10/CH02	DO-24V	Force output on / off	24 VDC on / off at terminals		

CAB-FGS-01 (FGS) - 12 channels

Tag	Address	Card	Test	Expected	Result	Init.
800-GD-101A	R1/S01/CH01	FGS-AI	Inject 4 / 12 / 20 mA	0 / 50 / 100 %LEL		
800-GD-111A	R1/S01/CH02	FGS-AI	Inject 4 / 12 / 20 mA	0 / 10 / 20 ppm		
800-GD-101B	R1/S02/CH01	FGS-AI	Inject 4 / 12 / 20 mA	0 / 50 / 100 %LEL		
800-GD-111B	R1/S02/CH02	FGS-AI	Inject 4 / 12 / 20 mA	0 / 10 / 20 ppm		
800-GD-111C	R1/S03/CH02	FGS-AI	Inject 4 / 12 / 20 mA	0 / 10 / 20 ppm		
800-FD-121A	R1/S04/CH01	FGS-DI	Simulate contact / NAMUR state	State change at HMI		
800-HS-131	R1/S04/CH02	FGS-DI	Simulate contact / NAMUR state	State change at HMI		
800-FD-121B	R1/S05/CH01	FGS-DI	Simulate contact / NAMUR state	State change at HMI		
800-HS-132	R1/S05/CH02	FGS-DI	Simulate contact / NAMUR state	State change at HMI		
800-FD-121C	R1/S06/CH01	FGS-DI	Simulate contact / NAMUR state	State change at HMI		
800-UA-141	R1/S07/CH01	FGS-DO	Force output on / off	24 VDC on / off at terminals		
800-UA-142	R1/S07/CH02	FGS-DO	Force output on / off	24 VDC on / off at terminals		

CAB-SIS-01 (SIS) - 56 channels

Tag	Address	Card	Test	Expected	Result	Init.
100-PT-011A	R1/S01/CH01	SIS-AI-IS	Inject 4 / 12 / 20 mA	0 / 75 / 150 barg		
200-PT-031A	R1/S01/CH02	SIS-AI-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 barg		
100-LT-013	R1/S01/CH03	SIS-AI-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 %		
300-PT-040A	R1/S01/CH04	SIS-AI-IS	Inject 4 / 12 / 20 mA	0 / 100 / 200 barg		
300-PT-044A	R1/S01/CH05	SIS-AI-IS	Inject 4 / 12 / 20 mA	0 / 5 / 10 barg		
400-PT-025A	R1/S01/CH06	SIS-AI-IS	Inject 4 / 12 / 20 mA	0 / 5 / 10 barg		
300-PT-040B	R1/S01/CH07	SIS-AI-IS	Inject 4 / 12 / 20 mA	0 / 100 / 200 barg		
100-PT-011B	R1/S02/CH01	SIS-AI-IS	Inject 4 / 12 / 20 mA	0 / 75 / 150 barg		
200-PT-031B	R1/S02/CH02	SIS-AI-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 barg		
300-PT-044B	R1/S02/CH05	SIS-AI-IS	Inject 4 / 12 / 20 mA	0 / 5 / 10 barg		
400-PT-025B	R1/S02/CH06	SIS-AI-IS	Inject 4 / 12 / 20 mA	0 / 5 / 10 barg		
100-PT-011C	R1/S03/CH01	SIS-AI-IS	Inject 4 / 12 / 20 mA	0 / 75 / 150 barg		
200-LT-041	R1/S03/CH02	SIS-AI-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 %		
400-TT-026	R1/S03/CH03	SIS-AI-IS	Inject 4 / 12 / 20 mA	0 / 125 / 250 degC		
700-LT-002	R1/S03/CH04	SIS-AI-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 %		
600-PT-005	R1/S03/CH05	SIS-AI-IS	Inject 4 / 12 / 20 mA	0 / 3 / 6 barg		
800-GD-101C	R1/S03/CH06	SIS-AI-IS	Inject 4 / 12 / 20 mA	0 / 50 / 100 %LEL		
100-ZSC-012A	R1/S04/CH01	SIS-DI-IS	Simulate contact / NAMUR state	State change at HMI		
100-ZSC-001	R1/S04/CH02	SIS-DI-IS	Simulate contact / NAMUR state	State change at HMI		
200-ZSC-001	R1/S04/CH03	SIS-DI-IS	Simulate contact / NAMUR state	State change at HMI		
200-ZSC-042	R1/S04/CH04	SIS-DI-IS	Simulate contact / NAMUR state	State change at HMI		
900-HS-002	R1/S04/CH05	SIS-DI-IS	Simulate contact / NAMUR state	State change at HMI		
300-ZSC-041	R1/S04/CH06	SIS-DI-IS	Simulate contact / NAMUR state	State change at HMI		
300-ZSC-042	R1/S04/CH07	SIS-DI-IS	Simulate contact / NAMUR state	State change at HMI		
400-ZSC-021	R1/S04/CH08	SIS-DI-IS	Simulate contact / NAMUR state	State change at HMI		
400-ZSC-022	R1/S04/CH09	SIS-DI-IS	Simulate contact / NAMUR state	State change at HMI		
400-ZSO-023	R1/S04/CH10	SIS-DI-IS	Simulate contact / NAMUR state	State change at HMI		
400-ZSC-024	R1/S04/CH11	SIS-DI-IS	Simulate contact / NAMUR state	State change at HMI		
400-HS-030	R1/S04/CH12	SIS-DI-IS	Simulate contact / NAMUR state	State change at HMI		
700-ZSO-011	R1/S04/CH13	SIS-DI-IS	Simulate contact / NAMUR state	State change at HMI		
700-ZSO-012	R1/S04/CH14	SIS-DI-IS	Simulate contact /	State change at HMI		

Tag	Address	Card	Test	Expected	Result	Init.
			NAMUR state			
700-ZSO-013	R1/S04/CH15	SIS-DI-IS	Simulate contact / NAMUR state	State change at HMI		
900-HS-001	R1/S05/CH01	SIS-DI	Simulate contact / NAMUR state	State change at HMI		
300-XS-035	R1/S05/CH02	SIS-DI	Simulate contact / NAMUR state	State change at HMI		
400-BE-020A	R1/S05/CH03	SIS-DI	Simulate contact / NAMUR state	State change at HMI		
900-HS-003	R1/S05/CH04	SIS-DI	Simulate contact / NAMUR state	State change at HMI		
400-BE-020B	R1/S05/CH05	SIS-DI	Simulate contact / NAMUR state	State change at HMI		
100-XV-012A	R1/S06/CH01	SIS-DO	Force output on / off	24 VDC on / off at terminals		
100-ESDV-001	R1/S06/CH02	SIS-DO	Force output on / off	24 VDC on / off at terminals		
200-ESDV-001	R1/S06/CH03	SIS-DO	Force output on / off	24 VDC on / off at terminals		
200-SDV-042	R1/S06/CH04	SIS-DO	Force output on / off	24 VDC on / off at terminals		
900-UA-003	R1/S06/CH05	SIS-DO	Force output on / off	24 VDC on / off at terminals		
300-ESDV-041	R1/S06/CH06	SIS-DO	Force output on / off	24 VDC on / off at terminals		
300-ESDV-042	R1/S06/CH07	SIS-DO	Force output on / off	24 VDC on / off at terminals		
400-XV-021	R1/S06/CH08	SIS-DO	Force output on / off	24 VDC on / off at terminals		
100-XV-012B	R1/S07/CH01	SIS-DO	Force output on / off	24 VDC on / off at terminals		
300-XY-043	R1/S07/CH02	SIS-DO	Force output on / off	24 VDC on / off at terminals		
400-XV-022	R1/S07/CH03	SIS-DO	Force output on / off	24 VDC on / off at terminals		
400-XV-023	R1/S07/CH04	SIS-DO	Force output on / off	24 VDC on / off at terminals		
400-XV-024	R1/S07/CH05	SIS-DO	Force output on / off	24 VDC on / off at terminals		
700-BDV-011	R1/S07/CH06	SIS-DO	Force output on / off	24 VDC on / off at terminals		
700-BDV-012	R1/S07/CH07	SIS-DO	Force output on / off	24 VDC on / off at terminals		
700-BDV-013	R1/S07/CH08	SIS-DO	Force output on / off	24 VDC on / off at terminals		
100-ZSC-012B	R1/S08/CH01	SIS-DI-IS	Simulate contact / NAMUR state	State change at HMI		
600-XY-001A	R1/S10/CH01	SIS-DO	Force output on / off	24 VDC on / off at terminals		
600-XY-001B	R1/S11/CH01	SIS-DO	Force output on / off	24 VDC on / off at terminals		

Spare channels

Cabinet	Spare channels listed	Check	Result	Init.
CAB-BPCS-01	27	Spare terminals wired, labelled, channel available		
CAB-BPCS-02	3	Spare terminals wired, labelled, channel available		
CAB-BPCS-03	35	Spare terminals wired, labelled, channel available		
CAB-BPCS-04	18	Spare terminals wired, labelled, channel available		
CAB-BPCS-05	68	Spare terminals wired, labelled, channel available		
CAB-FGS-01	44	Spare terminals wired, labelled, channel available		
CAB-SIS-01	40	Spare terminals wired, labelled, channel available		

8 Cause & effect functional tests

Each cause group is tested against the ESD cause & effect matrix, including the initiator voting. M-1 initiators must not trip; M initiators must trip.

Test	SIF / group	SIL	Voting	Initiators	Steps
CE-01	SIF-900-01	3	2oo3	100-PT-011A, 100-PT-011B, 100-PT-011C	14
CE-02	SIF-900-02	1	1oo1	100-LT-013	4
CE-03	SIF-900-03	2	1oo2	200-PT-031A, 200-PT-031B	6
CE-04	SIF-900-04	2	1oo1	200-LT-041	4
CE-05	SIF-900-05	1	1oo1	200-LT-031	4
CE-06	SIF-900-06	2	1oo2	300-PT-040A, 300-PT-040B	6

Test	SIF / group	SIL	Voting	Initiators	Steps
CE-07	SIF-900-07	1	2oo2	300-PT-044A, 300-PT-044B	8
CE-08	SIF-900-08	1	1oo1	300-XS-035	4
CE-09	SIF-900-09	1	1oo2	400-BE-020A, 400-BE-020B	6
CE-10	SIF-900-10	1	1oo2	400-PT-025A, 400-PT-025B	6
CE-11	SIF-900-11	1	1oo2	400-PT-025A, 400-PT-025B	6
CE-12	SIF-900-12	1	1oo1	400-TT-026	4
CE-13	manual	-	1oo1	400-HS-030	3
CE-14	SIF-900-13	1	1oo1	700-LT-002	4
CE-15	SIF-900-14	1	1oo1	600-PT-005	4
CE-16	manual	-	1oo1	900-HS-001	3
CE-17	manual	-	1oo1	900-HS-002	3
CE-18	manual	-	1oo1	900-HS-003	3
CE-19	FG-300-GAS-L	-	1oo3	800-GD-101A, 800-GD-101B, 800-GD-101C	8
CE-20	FG-300-GAS-H	-	2oo3	800-GD-101A, 800-GD-101B, 800-GD-101C	14
CE-21	FG-100-H2S-L	-	1oo3	800-GD-111A, 800-GD-111B, 800-GD-111C	8
CE-22	FG-100-H2S-H	-	2oo3	800-GD-111A, 800-GD-111B	8
CE-23	FG-300-FIRE	-	1oo3	800-FD-121A, 800-FD-121B, 800-FD-121C	8
CE-24	manual	-	1oo1	800-HS-131	3
CE-25	manual	-	1oo1	800-HS-132	3

CE-01 - HIPPS - inlet overpressure protection of unit 100/200

SIF: SIF-900-01 | SIL: 3 | Voting: 2oo3 | Initiators: 100-PT-011A, 100-PT-011B, 100-PT-011C

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Drive 100-PT-011A above the trip setpoint 120 barg; keep the other initiator(s) normal.	No action (1 of 3 initiators); alarm / discrepancy indication only.		
03	Return initiators to normal; reset from the SIS.	No change.		
04	Drive 100-PT-011B above the trip setpoint 120 barg; keep the other initiator(s) normal.	No action (1 of 3 initiators); alarm / discrepancy indication only.		
05	Return initiators to normal; reset from the SIS.	No change.		
06	Drive 100-PT-011C above the trip setpoint 120 barg; keep the other initiator(s) normal.	No action (1 of 3 initiators); alarm / discrepancy indication only.		
07	Return initiators to normal; reset from the SIS.	No change.		
08	Drive 100-PT-011A, 100-PT-011B above the trip setpoint 120 barg; keep the other initiator(s) normal.	SIS trips. Effects: 100-XV-012A close; 100-XV-012B close; 900-UA-003 energise.		
09	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
10	Drive 100-PT-011A, 100-PT-011C above the trip setpoint 120 barg; keep the other initiator(s) normal.	SIS trips. Effects: 100-XV-012A close; 100-XV-012B close; 900-UA-003 energise.		
11	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
12	Drive 100-PT-011B, 100-PT-011C above the trip setpoint 120 barg; keep the other initiator(s) normal.	SIS trips. Effects: 100-XV-012A close; 100-XV-012B close; 900-UA-003 energise.		
13	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
14	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-02 - Slug catcher high-high level - close inlet ESDV

SIF: SIF-900-02 | SIL: 1 | Voting: 1oo1 | Initiators: 100-LT-013

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Drive 100-LT-013 above the trip setpoint 85 %.	SIS trips. Effects: 100-ESDV-001 close; 900-UA-003 energise.		
03	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
04	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-03 - Separator high-high pressure - close separator inlet ESDV

SIF: SIF-900-03 | SIL: 2 | Voting: 1oo2 | Initiators: 200-PT-031A, 200-PT-031B

DATA ISSUES: 200-BDV-043 - IC-017: C&E effect references 200-BDV-043, which is not in the instrument index

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Drive 200-PT-031A above the trip setpoint 85 barg; keep the other initiator(s) normal.	SIS trips. Effects: 200-ESDV-001 close; 900-UA-003 energise; 200-BDV-043 open.		
03	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
04	Drive 200-PT-031B above the trip setpoint 85 barg; keep the other initiator(s) normal.	SIS trips. Effects: 200-ESDV-001 close; 900-UA-003 energise; 200-BDV-043 open.		
05	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
06	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-04 - Separator low-low level (gas blow-by) - close condensate outlet SDV

SIF: SIF-900-04 | SIL: 2 | Voting: 1oo1 | Initiators: 200-LT-041

DATA ISSUES: SIF-900-04 - IC-020: SIL 2 SIF uses 1oo1 initiators (HFT 0); minimum HFT 1 required

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Drive 200-LT-041 below the trip setpoint 10 %.	SIS trips. Effects: 200-SDV-042 close; 900-UA-003 energise.		
03	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
04	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-05 - Separator high-high level - close separator inlet ESDV

SIF: SIF-900-05 | SIL: 1 | Voting: 1oo1 | Initiators: 200-LT-031

DATA ISSUES: 200-LT-031 - IC-014: System segregation: system: instrument 'SIS' vs io_point 'BPCS'

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Drive 200-LT-031 above the trip setpoint 90 %.	SIS trips. Effects: 200-ESDV-001 close; 900-UA-003 energise.		
03	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
04	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-06 - Compressor discharge high-high pressure - trip driver, close suction and discharge ESDVs

SIF: SIF-900-06 | SIL: 2 | Voting: 1oo2 | Initiators: 300-PT-040A, 300-PT-040B

DATA ISSUES: 300-PT-040A - IC-012: Trip setpoint for SIF-900-06 differs between documents: SIF register 150 barg, C&E 148 barg, alarm list 150 barg | 300-PT-040B - IC-012: Trip setpoint for SIF-900-06 differs between documents: SIF register 150 barg, C&E 148 barg, alarm list 150 barg | 300-PT-040B - IC-025: Redundant devices 300-PT-040A, 300-PT-040B share card CAB-SIS-01/R1/S01 | 300-PT-040B - IC-026: C&E cause for 300-PT-040B (SIF-900-06) does not action final element 300-ESDV-042

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Drive 300-PT-040A above the trip setpoint 148 barg; keep the other initiator(s) normal.	SIS trips. Effects: 900-UA-003 energise; 300-XY-043 trip; 300-ESDV-041 close; 300-ESDV-042 close.		
03	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and		

No.	Action	Expected result	Result	Init.
		reset is applied; effects remain in safe state until reset.		
04	Drive 300-PT-040B above the trip setpoint 148 barg; keep the other initiator(s) normal.	SIS trips. Effects: 900-UA-003 energise; 300-XY-043 trip; 300-ESDV-041 close.		
05	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
06	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-07 - Lube oil header low-low pressure - trip compressor driver

SIF: SIF-900-07 | SIL: 1 | Voting: 2oo2 | Initiators: 300-PT-044A, 300-PT-044B

DATA ISSUES: 300-PT-044A - IC-027: C&E voting 2oo2 differs from SIF register 1oo2 for SIF-900-07 | 300-PT-044B - IC-027: C&E voting 2oo2 differs from SIF register 1oo2 for SIF-900-07

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Drive 300-PT-044A below the trip setpoint 1.5 barg; keep the other initiator(s) normal.	No action (1 of 2 initiators); alarm / discrepancy indication only.		
03	Return initiators to normal; reset from the SIS.	No change.		
04	Drive 300-PT-044B below the trip setpoint 1.5 barg; keep the other initiator(s) normal.	No action (1 of 2 initiators); alarm / discrepancy indication only.		
05	Return initiators to normal; reset from the SIS.	No change.		
06	Drive 300-PT-044A, 300-PT-044B below the trip setpoint 1.5 barg.	SIS trips. Effects: 300-XY-043 trip.		
07	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
08	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-08 - Machinery protection trip (vibration / axial displacement danger) - trip driver

SIF: SIF-900-08 | SIL: 1 | Voting: 1oo1 | Initiators: 300-XS-035

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Activate 300-XS-035 (open the trip contact / simulate the trip condition).	SIS trips. Effects: 300-XY-043 trip.		
03	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
04	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-09 - Reboiler flame failure - close main fuel and pilot, open vent

SIF: SIF-900-09 | SIL: 1 | Voting: 1oo2 | Initiators: 400-BE-020A, 400-BE-020B

DATA ISSUES: 400-BE-020B - IC-022: certified temperature class T2 does not cover area T3 | 400-BE-020B - IC-025: Redundant devices 400-BE-020A, 400-BE-020B share card CAB-SIS-01/R1/S05 | 400-XV-023 - IC-032: C&E effect E12 action 'CLOSE' does not match the safe state 'OPEN' in the instrument index (expected 'OPEN')

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Activate 400-BE-020A (open the trip contact / simulate the trip condition); keep the other initiator(s) normal.	SIS trips. Effects: 400-XV-021 close; 400-XV-022 close; 400-XV-023 close; 400-XV-024 close.		
03	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
04	Activate 400-BE-020B (open the trip contact / simulate the trip condition); keep the other initiator(s) normal.	SIS trips. Effects: 400-XV-021 close; 400-XV-022 close; 400-XV-023 close; 400-XV-024 close.		
05	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
06	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-10 - Fuel gas supply pressure low-low - close main fuel double block, open vent

SIF: SIF-900-10 | SIL: 1 | Voting: 1oo2 | Initiators: 400-PT-025A, 400-PT-025B

DATA ISSUES: 400-XV-023 - IC-032: C&E effect E12 action 'CLOSE' does not match the safe state 'OPEN' in the instrument index (expected 'OPEN')

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Drive 400-PT-025A below the trip setpoint 1 barg; keep the other initiator(s) normal.	SIS trips. Effects: 400-XV-021 close; 400-XV-022 close; 400-XV-023 close.		
03	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
04	Drive 400-PT-025B below the trip setpoint 1 barg; keep the other initiator(s) normal.	SIS trips. Effects: 400-XV-021 close; 400-XV-022 close; 400-XV-023 close.		
05	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
06	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-11 - Fuel gas supply pressure high-high - close main fuel double block, open vent

SIF: SIF-900-11 | SIL: 1 | Voting: 1oo2 | Initiators: 400-PT-025A, 400-PT-025B

DATA ISSUES: 400-XV-023 - IC-032: C&E effect E12 action 'CLOSE' does not match the safe state 'OPEN' in the instrument index (expected 'OPEN')

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Drive 400-PT-025A above the trip setpoint 6 barg; keep the other initiator(s) normal.	SIS trips. Effects: 400-XV-021 close; 400-XV-022 close; 400-XV-023 close.		
03	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
04	Drive 400-PT-025B above the trip setpoint 6 barg; keep the other initiator(s) normal.	SIS trips. Effects: 400-XV-021 close; 400-XV-022 close; 400-XV-023 close.		
05	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
06	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-12 - Reboiler temperature high-high - close main fuel and pilot, open vent

SIF: SIF-900-12 | SIL: 1 | Voting: 1oo1 | Initiators: 400-TT-026

DATA ISSUES: 400-XV-023 - IC-032: C&E effect E12 action 'CLOSE' does not match the safe state 'OPEN' in the instrument index (expected 'OPEN')

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Drive 400-TT-026 above the trip setpoint 215 degC.	SIS trips. Effects: 400-XV-021 close; 400-XV-022 close; 400-XV-023 close; 400-XV-024 close.		
03	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
04	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-13 - Reboiler BMS local emergency stop

Manual initiation | Voting: 1oo1 | Initiators: 400-HS-030

DATA ISSUES: 400-XV-023 - IC-032: C&E effect E12 action 'CLOSE' does not match the safe state 'OPEN' in the instrument index (expected 'OPEN')

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Operate 400-HS-030 (Reboiler BMS local emergency stop).	Effects: 400-XV-021 close; 400-XV-022 close; 400-XV-023 close; 400-XV-024 close.		
03	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-14 - Flare KO drum high-high level - close inlet and separator ESDVs

SIF: SIF-900-13 | SIL: 1 | Voting: 1oo1 | Initiators: 700-LT-002

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Drive 700-LT-002 above the trip setpoint 85 %.	SIS trips. Effects: 100-ESDV-001 close; 200-ESDV-001 close; 900-UA-003 energise.		
03	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
04	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-15 - Condensate export pump suction low-low pressure - trip pumps

SIF: SIF-900-14 | SIL: 1 | Voting: 1oo1 | Initiators: 600-PT-005

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Drive 600-PT-005 below the trip setpoint 1 barg.	SIS trips. Effects: 600-XY-001A trip; 600-XY-001B trip.		
03	Return initiators to normal; reset from the SIS.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
04	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-16 - ESD-1 manual (control room)

Manual initiation | Voting: 1oo1 | Initiators: 900-HS-001

DATA ISSUES: 400-XV-023 - IC-032: C&E effect E12 action 'CLOSE' does not match the safe state 'OPEN' in the instrument index (expected 'OPEN')

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Operate 900-HS-001 (ESD-1 manual (control room)).	Effects: 100-XV-012A close; 100-XV-012B close; 100-ESDV-001 close; 200-ESDV-001 close; 200-SDV-042 close; 900-UA-003 energise; 300-XY-043 trip; 300-ESDV-041 close; 300-ESDV-042 close; 400-XV-021 close; 400-XV-022 close; 400-XV-023 close; 400-XV-024 close; 600-XY-001A trip; 600-XY-001B trip.		
03	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-17 - ESD-1 manual (field)

Manual initiation | Voting: 1oo1 | Initiators: 900-HS-002

DATA ISSUES: 400-XV-023 - IC-032: C&E effect E12 action 'CLOSE' does not match the safe state 'OPEN' in the instrument index (expected 'OPEN')

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Operate 900-HS-002 (ESD-1 manual (field)).	Effects: 100-XV-012A close; 100-XV-012B close; 100-ESDV-001 close; 200-ESDV-001 close; 200-SDV-042 close; 900-UA-003 energise; 300-XY-043 trip; 300-ESDV-041 close; 300-ESDV-042 close; 400-XV-021 close; 400-XV-022 close; 400-XV-023 close; 400-XV-024 close; 600-XY-001A trip; 600-XY-001B		

No.	Action	Expected result	Result	Init.
		trip.		
03	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-18 - Emergency depressurisation (control room)

Manual initiation | Voting: 1oo1 | Initiators: 900-HS-003

DATA ISSUES: 700-BDV-012 - IC-032: C&E effect E15 action 'OPEN' does not match the safe state 'CLOSED' in the instrument index (expected 'CLOSE')

No.	Action	Expected result	Result	Init.
01	Confirm SIS in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Operate 900-HS-003 (Emergency depressurisation (control room)).	Effects: 900-UA-003 energise; 700-BDV-011 open; 700-BDV-012 open; 700-BDV-013 open.		
03	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-19 - FG-300-GAS-L - HC gas low, compressor house (F&G)

Voting group: FG-300-GAS-L | Voting: 1oo3 | Initiators: 800-GD-101A, 800-GD-101B, 800-GD-101C

DATA ISSUES: 800-GD-101B - IC-034: H setpoint 60 %LEL differs from the standard 50 %LEL for GD | 800-GD-101C - IC-014: System segregation: system: instrument 'FGS' vs io_point 'SIS'

No.	Action	Expected result	Result	Init.
01	Confirm F&G system in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Drive 800-GD-101A above the setpoint 20 %LEL; keep the other initiator(s) normal.	F&G system activates. Effects: 800-UA-142 energise.		
03	Return initiators to normal; reset from the F&G system.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
04	Drive 800-GD-101B above the setpoint 20 %LEL; keep the other initiator(s) normal.	F&G system activates. Effects: 800-UA-142 energise.		
05	Return initiators to normal; reset from the F&G system.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
06	Drive 800-GD-101C above the setpoint 20 %LEL; keep the other initiator(s) normal.	F&G system activates. Effects: 800-UA-142 energise.		
07	Return initiators to normal; reset from the F&G system.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
08	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-20 - FG-300-GAS-H - HC gas high, compressor house (F&G)

Voting group: FG-300-GAS-H | Voting: 2oo3 | Initiators: 800-GD-101A, 800-GD-101B, 800-GD-101C

DATA ISSUES: 800-GD-101B - IC-034: H setpoint 60 %LEL differs from the standard 50 %LEL for GD | 800-GD-101C - IC-014: System segregation: system: instrument 'FGS' vs io_point 'SIS'

No.	Action	Expected result	Result	Init.
01	Confirm F&G system in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Drive 800-GD-101A above the setpoint 50 %LEL; keep the other initiator(s) normal.	No action (1 of 3 initiators); alarm / discrepancy indication only.		
03	Return initiators to normal; reset from the F&G system.	No change.		
04	Drive 800-GD-101B above the setpoint 50 %LEL; keep the other initiator(s) normal.	No action (1 of 3 initiators); alarm / discrepancy indication only.		
05	Return initiators to normal; reset from the F&G system.	No change.		
06	Drive 800-GD-101C above the setpoint 50 %LEL; keep the other initiator(s) normal.	No action (1 of 3 initiators); alarm / discrepancy indication only.		
07	Return initiators to normal; reset from the F&G system.	No change.		
08	Drive 800-GD-101A, 800-GD-101B above the setpoint 50 %LEL; keep the other initiator(s) normal.	F&G system activates. Effects: 800-UA-142 energise; 300-XY-043 trip; 300-ESDV-041 close; 300-ESDV-042 close.		
09	Return initiators to normal; reset from the F&G system.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until		

No.	Action	Expected result	Result	Init.
		reset.		
10	Drive 800-GD-101A, 800-GD-101C above the setpoint 50 %LEL; keep the other initiator(s) normal.	F&G system activates. Effects: 800-UA-142 energise; 300-XY-043 trip; 300-ESDV-041 close; 300-ESDV-042 close.		
11	Return initiators to normal; reset from the F&G system.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
12	Drive 800-GD-101B, 800-GD-101C above the setpoint 50 %LEL; keep the other initiator(s) normal.	F&G system activates. Effects: 800-UA-142 energise; 300-XY-043 trip; 300-ESDV-041 close; 300-ESDV-042 close.		
13	Return initiators to normal; reset from the F&G system.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
14	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-21 - FG-100-H2S-L - H2S low, inlet facilities (F&G)

Voting group: FG-100-H2S-L | Voting: 1oo3 | Initiators: 800-GD-111A, 800-GD-111B, 800-GD-111C

No.	Action	Expected result	Result	Init.
01	Confirm F&G system in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Drive 800-GD-111A above the setpoint 5 ppm; keep the other initiator(s) normal.	F&G system activates. Effects: 800-UA-142 energise.		
03	Return initiators to normal; reset from the F&G system.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
04	Drive 800-GD-111B above the setpoint 5 ppm; keep the other initiator(s) normal.	F&G system activates. Effects: 800-UA-142 energise.		
05	Return initiators to normal; reset from the F&G system.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
06	Drive 800-GD-111C above the setpoint 5 ppm; keep the other initiator(s) normal.	F&G system activates. Effects: 800-UA-142 energise.		
07	Return initiators to normal; reset from the F&G system.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
08	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-22 - FG-100-H2S-H - H2S high, inlet facilities (F&G)

Voting group: FG-100-H2S-H | Voting: 2oo3 | Initiators: 800-GD-111A, 800-GD-111B

DATA ISSUES: FG-100-H2S-H - IC-036: Voting 2oo3 but only 2 initiator(s) listed in the F&G C&E

No.	Action	Expected result	Result	Init.
01	Confirm F&G system in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Drive 800-GD-111A above the setpoint 10 ppm; keep the other initiator(s) normal.	No action (1 of 3 initiators); alarm / discrepancy indication only.		
03	Return initiators to normal; reset from the F&G system.	No change.		
04	Drive 800-GD-111B above the setpoint 10 ppm; keep the other initiator(s) normal.	No action (1 of 3 initiators); alarm / discrepancy indication only.		
05	Return initiators to normal; reset from the F&G system.	No change.		
06	Drive 800-GD-111A, 800-GD-111B above the setpoint 10 ppm.	F&G system activates. Effects: 800-UA-142 energise; 100-ESDV-001 close.		
07	Return initiators to normal; reset from the F&G system.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
08	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-23 - FG-300-FIRE - Fire detected, compressor house (F&G)

Voting group: FG-300-FIRE | Voting: 1oo3 | Initiators: 800-FD-121A, 800-FD-121B, 800-FD-121C

DATA ISSUES: FG-300-FIRE - IC-035: Voting 1oo3 drives executive actions (300-ESDV-041, 300-ESDV-042, 300-XY-043, 700-BDV-013); at least 2ooN required

No.	Action	Expected result	Result	Init.
01	Confirm F&G system in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Activate 800-FD-121A (open the trip contact / simulate the trip condition); keep the other initiator(s) normal.	F&G system activates. Effects: 800-UA-141 energise; 300-XY-043 trip; 300-ESDV-041 close; 300-ESDV-042 close; 700-BDV-013 open.		
03	Return initiators to normal; reset from the F&G system.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
04	Activate 800-FD-121B (open the trip contact / simulate the trip condition); keep the other initiator(s) normal.	F&G system activates. Effects: 800-UA-141 energise; 300-XY-043 trip; 300-ESDV-041 close; 300-ESDV-042 close; 700-BDV-013 open.		
05	Return initiators to normal; reset from the F&G system.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
06	Activate 800-FD-121C (open the trip contact / simulate the trip condition); keep the other initiator(s) normal.	F&G system activates. Effects: 800-UA-141 energise; 300-XY-043 trip; 300-ESDV-041 close; 300-ESDV-042 close; 700-BDV-013 open.		
07	Return initiators to normal; reset from the F&G system.	Trip resets only after all initiators are normal and reset is applied; effects remain in safe state until reset.		
08	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-24 - Manual call point, compressor house

Manual initiation | Voting: 1oo1 | Initiators: 800-HS-131

No.	Action	Expected result	Result	Init.
01	Confirm F&G system in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Operate 800-HS-131 (Manual call point, compressor house).	Effects: 800-UA-141 energise.		
03	Restore all initiators and effects; remove test equipment.	System back to normal.		

CE-25 - Manual call point, inlet facilities

Manual initiation | Voting: 1oo1 | Initiators: 800-HS-132

No.	Action	Expected result	Result	Init.
01	Confirm F&G system in test mode, all initiators healthy, all effects reset and in the normal state; bypasses removed.	No active trips, alarms or bypasses.		
02	Operate 800-HS-132 (Manual call point, inlet facilities).	Effects: 800-UA-141 energise.		
03	Restore all initiators and effects; remove test equipment.	System back to normal.		

9 Sequence tests (burner management)

Each sequence is stepped through with every permissive checked, step times recorded, and the failure case for the critical steps demonstrated.

SEQ-01 - BMS-400

DATA ISSUES: 400-BE-020B - IC-022: certified temperature class T2 does not cover area T3 | 400-BE-020B - IC-025: Redundant devices 400-BE-020A, 400-BE-020B share card CAB-SIS-01/R1/S05 | 400-XV-023 - IC-032: C&E effect E12 action 'CLOSE' does not match the safe state 'OPEN' in the instrument index (expected 'OPEN') | 400-ZSC-204 - IC-030: Sequence step references 400-ZSC-204, which is not in the instrument index | BMS-400 step 01 - IC-031: PURGE time 120 s is below the minimum 300 s | BMS-400 step 03 - IC-031: PILOT_TFI time 15 s exceeds the maximum 10 s

No.	Action	Expected result	Result	Init.
01	Confirm BMS-400 is reset, fuel and pilot valves closed, vent open, no flame detected.	BMS in 'ready to purge' state; no lock-out.		
02	Step 01 PURGE: Pre-ignition purge of the reboiler firebox. Check: 400-ZSC-021 = CLOSED; 400-ZSC-022 = CLOSED; 400-ZSO-023 = OPEN; 400-ZSC-204 = CLOSED; 400-BE-020A = NO FLAME; 400-BE-020B = NO FLAME.	Each condition shown met at the BMS HMI before the step proceeds.		
03	Time step 01 with a stopwatch / event log.	Step duration 120 s as configured (record the measured time).		
04	Negative test, step 01: Start the purge with one permissive not met (e.g. a fuel valve not proven closed).	Purge does not start; purge timer does not run.		

No.	Action	Expected result	Result	Init.
05	Step 02 VALVE_PROVING: Fuel gas valve proving - pressure hold between the block valves. Check: 400-PT-012 = PRESSURE STABLE.	Each condition shown met at the BMS HMI before the step proceeds.		
06	Time step 02 with a stopwatch / event log.	Step duration 30 s as configured (record the measured time).		
07	Negative test, step 02: Simulate a pressure rise / decay between the block valves during the hold.	Valve proving fails; BMS locks out; light-off is not permitted.		
08	Step 03 PILOT_TFI: Pilot trial for ignition - open pilot valve, energise igniter. Check: 400-XV-024 = OPEN.	Each condition shown met at the BMS HMI before the step proceeds.		
09	Time step 03 with a stopwatch / event log.	Step duration 15 s as configured (record the measured time).		
10	Negative test, step 03: Inhibit the pilot flame (scanners see no flame) for the whole trial.	At the end of the trial the pilot valve closes and the BMS locks out.		
11	Step 04 PILOT_PROVE: Pilot flame proven on both scanners. Check: 400-BE-020A = FLAME; 400-BE-020B = FLAME.	Each condition shown met at the BMS HMI before the step proceeds.		
12	Negative test, step 04: Remove the flame signal of one scanner after pilot proving.	Flame failure trip: main fuel and pilot close, vent opens; lock-out.		
13	Step 05 MAIN_TFI: Main flame trial for ignition - open double block, close vent. Check: 400-XV-021 = OPEN; 400-XV-022 = OPEN; 400-XV-023 = CLOSED.	Each condition shown met at the BMS HMI before the step proceeds.		
14	Time step 05 with a stopwatch / event log.	Step duration 10 s as configured (record the measured time).		
15	Negative test, step 05: Inhibit the main flame for the whole trial.	At the end of the trial the double block closes, vent opens; lock-out.		
16	Step 06 RUN: Release to reboiler temperature control. Check: 400-TV-010 = AUTO.	Each condition shown met at the BMS HMI before the step proceeds.		
17	Reset the BMS and restore the normal state.	System back to normal.		

10 Test philosophy and acceptance criteria

Placeholder - narrative not yet drafted
[NARRATIVE TO BE DRAFTED - acceptance] Pass/fail criteria, tolerance, handling of failures and punch items.

11 Punch list and deviations

No.	Test / item	Description	Category	Raised by	Closed
1					
2					
3					
4					
5					
6					
7					
8					

12 Test completion

FAT completed with all tests passed or punch items recorded above.

Role	Company	Name	Signature	Date
Vendor				
Contractor				
Company witness				

Generation record

Item	Value
Status	DRAFT - not checked
Generated	2026-10-10T15:24:45+00:00
Engine	iccopilot 0.1.0
Template	fat_procedure v1.1 (sha 3d4e2b7bbc52)
Data snapshot	0b4feb1c32b70824
Consistency run	20261010T152443Z-4de462
Requested by	K. Berhane
Narrative	none - placeholders left for the engineer